Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India
(An Autonomous Institute Affiliated to University of Mumbai)

End Semester Examination

Nov. 2017

Max. Marks: 100
Duration: 180 Minutes
Class: T. E. I. T
Semester: V
Course Code: ITC 502
Branch: IT
Name of the Course: Operating System

Instructions:

(1) All Questions are Compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Question No.		Max. Marks	CO
Q 1 (a)	Operating System is known as a "Resource Manager." Explain your justifications with examples.	8	1
Q 1 (b)	Discuss the need of Bootstrap Process.	2	1
Q 1 (c)	What is a "Kernel"? Differentiate between Monolithic and Microkernel.	4	1
Q 1 (d)	Discuss the importance of "System Calls." Write a sudo code to explain "fork()" system call.	6	1
Q2 (a)	Draw process state diagram and explain the following states: 1. New 2. Ready 3. Running 4. Wait 5. Suspended ready 6. Suspended wait	4	2
Q2 (b)	Find Average Waiting Time for the following processes using Shortest Job First (Non Pre-emptive). Draw proper gant chart also.	8	2

Process No	Arrival Time(ms)	Burst Time(ms)
P1	1	5
P2	2	8
P3	3	6
P4	4	2
P5	5	4

<table>
<tr><td></td><td>OR

Explain Short-Term, Medium –Term and Long –Term Scheduler with a diagram.</td><td></td><td>2</td></tr>
<tr><td>Q 2 (c)</td><td>Analyze the impact of Context Switching on Average Turn Around Time using Round Robin Scheduling with time quantum as 3ms, 6ms and 10ms for the following processes.

<table>
<tr><td>Process No</td><td>Arrival Time (ms)</td><td>Burst Time (ms)</td></tr>
<tr><td>P1</td><td>0</td><td>4</td></tr>
<tr><td>P2</td><td>1</td><td>8</td></tr>
<tr><td>P3</td><td>2</td><td>6</td></tr>
<tr><td>P4</td><td>3</td><td>15</td></tr>
<tr><td>P5</td><td>4</td><td>10</td></tr>
</table></td><td>8</td><td>3</td></tr>
<tr><td>Q3 (a)</td><td>Consider a disk with 200 tracks numbered from 0 to 199. The following is the sequence of disk track requests: 27,129,110,186,147,41,10,64,120. Assume that disk head is initially positioned over 100 and is moving in the direction of decreasing track number. Perform the following algorithms.
a) SSTF b) C-SCAN
Calculate the average seek length.</td><td>8</td><td>2</td></tr>
<tr><td>Q3 (b)</td><td>Explain the file allocation techniques.</td><td>6</td><td>5</td></tr>
<tr><td>Q 3 (c)</td><td>Discuss Paging in detail with a neat diagram.</td><td>6</td><td>2</td></tr>
<tr><td>Q4 (a)</td><td>What example can you find to demonstrate "Race Condition"? Producer-Consumer problem can be solved with sleep-wake up calls. What other way you can use to solve this problem? Write a sudo code.

OR

Describe Critical Section and discuss the necessary and sufficient conditions to solve critical section problem. Differentiate between binary and counting semaphores.</td><td>7</td><td>3

3</td></tr>
<tr><td>Q4 (b)</td><td>Use Resource –Allocation graph to explain deadlock. Illustrate necessary and sufficient conditions for deadlock.</td><td>5</td><td>3</td></tr>
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Q 4 (c)	How would you apply Bankers algorithm to avoid having deadlock in the following situation? Available resources are A=3, B=3 and C=2 [Marks will be given on every step] Write the safe state if any.	8	3
Q5 (a)	What is dynamic partitioning? Explain the technique to overcome the disadvantage of it.	6	4
Q5 (b)	Design and describe Inode structure of UNIX OS. Explain with an example.	6	4
Q 5 (c)	Calculate the hit and miss ratio for the following string using page replacement policies- LRU, OPTIMAL. Compare it with frame size 3 and 4. 1,2,3,2,1,5,2,1,6,2,5,6,3,1,3,6,1,2,4,3 Explain the effect of page size on performance?	8	4

Table for Q 4 (c):

Process	Allocation			Maximum		
	A	B	C	A	B	C
P0	2	0	0	7	5	3
P1	0	1	0	3	2	2
P2	2	1	1	2	2	2
P3	3	0	2	9	0	2
P4	0	0	2	4	3	3